

PV ROHAN

+91 99209 13048 | rohan.051031@gmail.com | linkedin.com/in/pvrohan | github.com/pvrohan

Summary

Machine Learning Engineer and full-stack developer with hands-on experience building AI-driven systems, multi-agent platforms, and real-time data pipelines. Proficient in Python, TensorFlow, PyTorch, and LangChain; adept at deploying scalable REST APIs and integrating LLMs into production-ready applications. Seeking a challenging role to develop and ship cutting-edge ML solutions.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, Dart, SQL

AI / ML: TensorFlow, PyTorch, Scikit-learn, LangChain, Google Gemini, Prompt Engineering

Web & Mobile: React.js (Vite), FastAPI, Flutter, HTML/CSS, REST APIs, JWT Auth

Tools & Platforms: Docker, Git, GitHub, VS Code, Power BI, Postman, Twilio, Slack Webhooks

Projects

Pulse-Nexus | *Multi-Agent Post-Operative Monitoring System*

Dec 2025

- Designed and orchestrated specialised Google ADK agents — **Bio-Sentry** (vitals monitoring), **Data-Analyst** (trend detection), and **Clinical-Core** (guideline matching) — behind a unified `/run` REST API to deliver real-time risk scores, reasoning traces, and clinician-ready triage summaries for post-surgical patients.
- Implemented a multi-layered wearable data ingestion pipeline that cross-references live patient vitals (heart rate, SpO2, temperature) against clinical rule-sets, automatically escalating alerts when deterioration thresholds are breached.
- Exposed the agent network via a localhost API server; scripted 200/422 status tests and produced auto-generated API documentation via Postman Agent Mode, enabling rapid onboarding and contract-first development.
- Stack:** Google ADK (Python), REST APIs, Postman Agent Mode, JSON schema validation, Wearable data processing

Monacos | *Indoor Health Intelligence Hub*

Aug 2025 – Present

- Architected a full-stack IoT health platform ingesting real-time sensor streams (temperature, humidity, PM2.5, PM10, noise, light) from an ESP32 hub via Web Bluetooth and computing a 0–100 Indoor Health Score benchmarked against WHO and ASHRAE standards.
- Built a React (Vite) + TypeScript dashboard with Recharts visualisations, smart hazard alerts with colour-coded severity, and multi-room switching; developed a FastAPI + SQLite backend handling sensor ingestion, device metadata, and aggregated REST analytics.
- Integrated an AI *Health Guardian* (LangChain + Gemini 2 Flash) performing 7-day trend analysis, short-term air-quality and temperature forecasting, and room-specific improvement recommendations from historical sensor data.
- Secured the platform with JWT-based authentication, profile and avatar management, and a role-ready layout designed for multi-user and household expansion.
- Stack:** React (Vite), TypeScript, TailwindCSS, Shadcn UI, Recharts, FastAPI, SQLite, LangChain, Google Gemini 2 Flash, JWT Auth, Web Bluetooth, ESP32

InsurRenew Pro | *Bilingual AI Agent for Insurance Renewal & Upsell*

Oct 2025

- Engineered a production-grade bilingual (English/Hindi) conversational AI agent that automates the insurance renewal workflow — handling price objections, consultative upselling across policy tiers, and multi-turn intent management without a traditional knowledge base.
- Designed complex multi-stage conversational logic via a central System Prompt, enabling the agent to dynamically branch across renewal, objection-handling, and cross-sell flows while preserving natural dialogue continuity.
- Integrated live production APIs: **Twilio** for SMS payment links on policy confirmation and **Slack Webhooks** for context-rich human-agent handoff alerts; adopted a modular three-layer architecture (Prompt / Actions / Integrations) for independent iteration on logic, skills, and connectors.
- Stack:** Inya.ai (No-Code AI Platform), System Prompt Engineering, REST APIs, Twilio SMS, Slack Webhooks

Education

VIT Bhopal University

Bachelor of Technology, Computer Science and Engineering

2023 – 2027

Bhopal, India

Certifications

- Certified Ethical Hacker (CEH)** – EC-Council
- Introduction to Machine Learning** – NPTEL IIT Madras

Dec 2024

Apr 2025